

8. Acquisition D: detection trials

A detection trial is one frame at a fresh random lateral offset. All features and blank sites sit on one plate, so a single frame yields one trial for every feature at once. The detection levels are the (feature, station) pairs: 3 features (FEATURE_COUNT) at 9 stations give 27 values of the subtended size D_{px} , spaced by the station ratio $2^{(1/4)}$ (Z_STATION_RATIO), with each feature spanning two octaves and overlapping its neighbors by half an octave. The 10% and 5% points need many more trials than the 50% point, at the levels where detection is rare, which lie at the far stations. The 0% point is not measured: it is extrapolated from the fitted curve and reported as a prediction.

1. Post check. Run the quick-look detection count (Section 13, Step 2) on the first frame of each C pose at the reference station, 800 mm (Z_REFERENCE_MM). If any post-only control site shows detections above the false-alarm rate, re-make the disk posts thinner and repeat the check.
2. Main trials. For each configuration, disk or cutout at the small or the large gap, at every station of the ladder, capture 60 poses (DETECTION_TRIALS_PER_LEVEL), each at a new random lateral offset drawn as in Section 7, with 1 frame (FRAMES_PER_DETECTION_TRIAL).
3. Extended trials for the low point. At the 3 farthest stations, 1131, 1345, and 1600 mm (DETECTION_LOW_STATION_COUNT), where the smallest feature lies near and below the threshold, raise the pose count to 300 (DETECTION_LOW_TRIALS) for every configuration. Every feature and every blank site then has 300 trials there, which measures a detection probability of 5% (DETECTION_LOW_PROBABILITY) to about $\pm 2.5\%$ at 95% confidence and separates it from the 1% false-alarm rate. The 0% point is not measured: Section 13 extrapolates the fitted curve below the 5% point and reports the result as a prediction, clearly flagged.
4. Independence rule. Never use two frames from the same pose as separate trials; the fixed projector pattern and the target's fixed position make them strongly correlated. Only the first frame of a C pose may serve as a D trial (Step 5). Section 13, Step 1 checks independence after the fact.
5. Reuse. The first frame of each C pose at a matching configuration may count as a D trial, 30 of the 60 per configuration and station. The Section 9 budget assumes no reuse; the plan tool applies it on request and then plans 30 new D poses instead of 60 at the main stations.